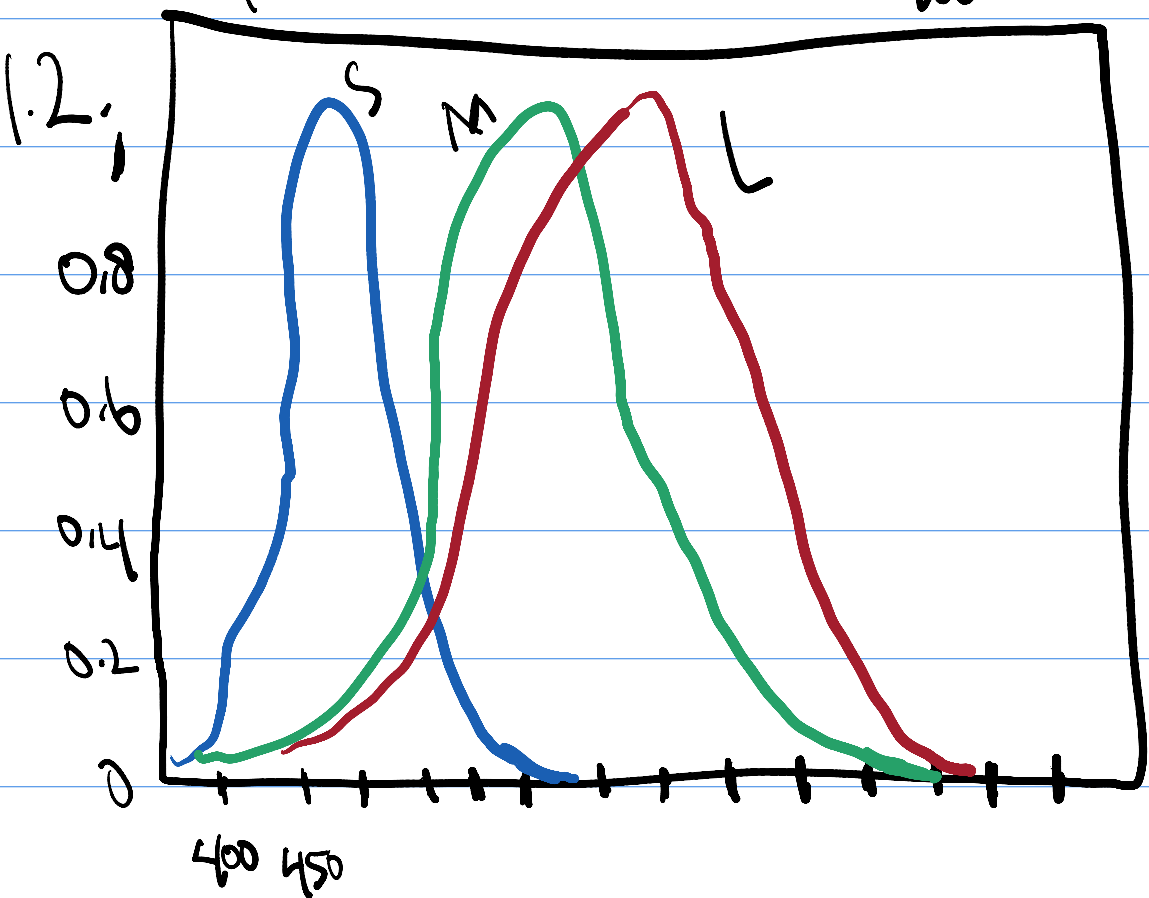
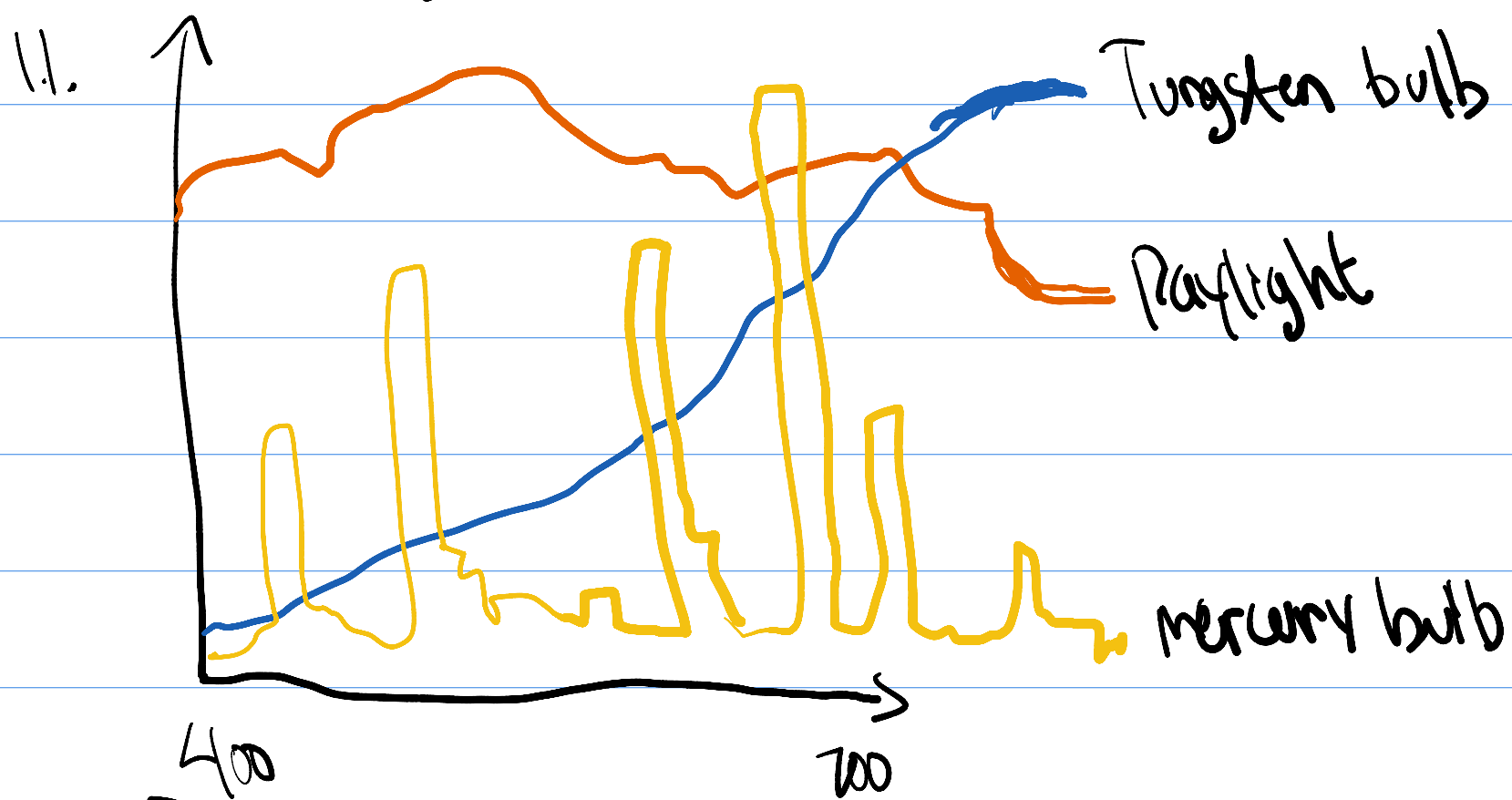


mmdb-ex03



1.3. Metamers is a color stimuli that has different wavelength composition but are perceived the same by the observer due to the trichromatic nature of vision

1.4. This phenomenon is called chromatic adaptation as human eyes ability to adjust to changes to illumination to preserve the appearance of object colors

2.1. Color model is an abstract way to define color that matches the system of human vision.

2.2. RGB = • Black, white | Primary (R, G, B) | Secondary (C, M, Y)
• Brightness produce colors (Additive)
• commonly used in monitors, TV

CMYK = • Black, white | Primary (Cyan, Magenta, Yellow, K (Black))
• Pigments and dyes produce colors (Subtractive)
• commonly used in printers

HSL = • Saturation (0 = gray, 1 = full color), Brightness (0 = black, 1 = white)
• Hue color (0 - 360°)
• commonly used in computer vision, digital design software

2.3. RGB = R : 0.2 CMYK / C = 1 - R = 0.8
 G : 0.6 M = 1 - G = 0.4

$$B = 0.3$$

$$\begin{cases} Y = 1 - B = 0.7 \\ K = \min(L, M, Y) = 0.14 \end{cases}$$

$$\text{HSV} \rightarrow W = \min(0.12, 0.6, 0.3) = 0.2$$

$$R' = R - W = 0 \quad | \quad G' = G - W = 0.4 \quad | \quad B' = B - W = 0.1$$

$$R' = 0, \text{ so } H = \left(0.1 \times \frac{120}{0.5}\right) + 120^\circ = 144^\circ$$

$$S = \frac{0.6 - 0.12}{0.6} = 0.667 = 66.7\%$$

$$V = 0.6 = 60\%$$

3.1 Color difference at 240p: Video is stored in YCbCr color space. When resolution is reduced, the video player may use incorrect conversion coefficient, causing color shift

3.2 CIE XYZ: Additive, linear light color space useful for calculating color mixtures but not perceptually similar.

CIE $L^*a^*b^*$: Oriented towards physiological properties of human color perception. Distance between colors accurately predicts perceived similarity